

Bird-Deterrent System for Medical Delivery Drones

A comprehensive solution for autonomous medical delivery drones operating in hilly terrain with hostile bird encounters.

Problem Statement: Design a system for a medicine delivery drone to deter bird attacks around the 5 km mark of a 10 km flight path in hilly terrain prone to drizzles and wind gusts.

Constraints:



- Cost under Rs 50,000
- Weight under 15 kg
- Must withstand drizzles and wind gusts

Our solution employs a multi-layered approach combining real-time bird detection and innovative deterrent mechanisms, allowing the drone to complete its critical medical delivery mission safely and reliably.

System Design: Layered Architecture

Sensor Layer

Forward-facing camera captures video feed
for bird detection

Additional sensors for navigation and
obstacle avoidance

Control Layer

Autonomous operation with remote override
capability

LoRa or 4G/LTE for long-range
communication

Failsafe protocols for various scenarios



Processing Layer

Raspberry Pi 4 with YOLOv4-tiny AI model

Real-time bird detection and classification

Optional Coral TPU accelerator for improved
performance

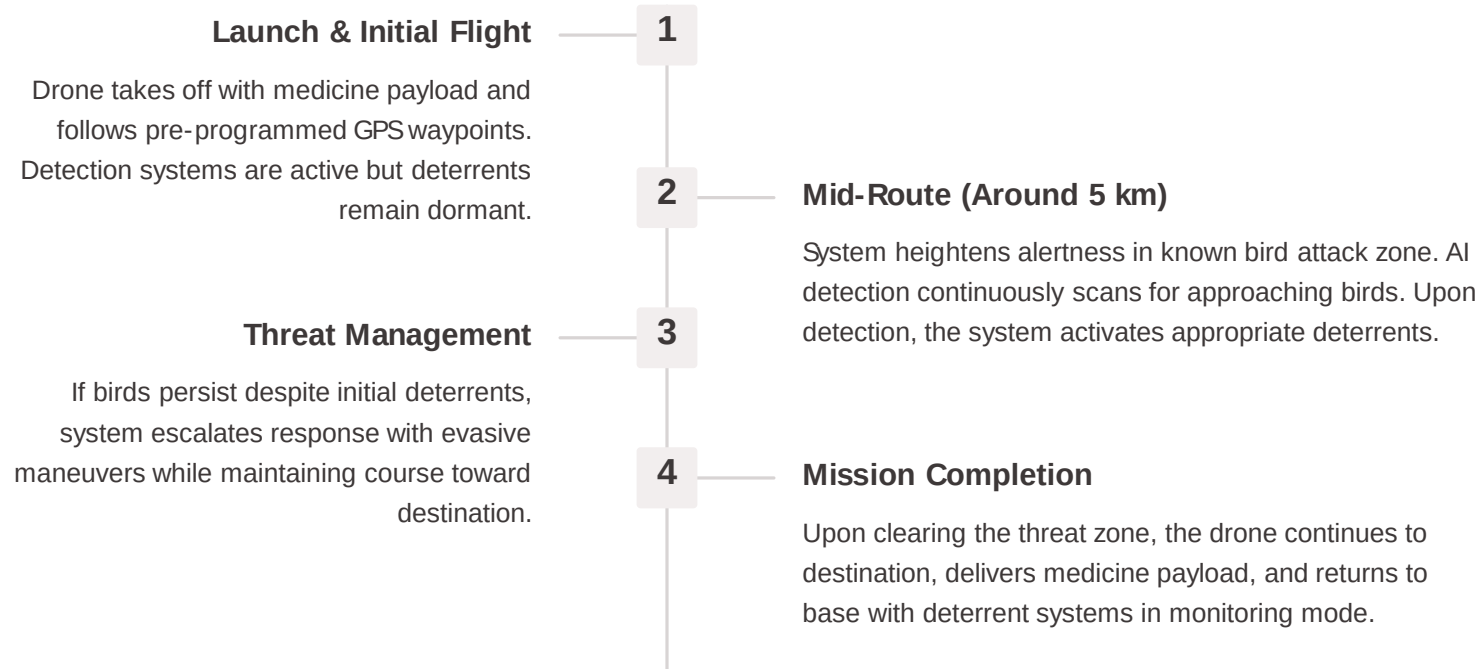
Deterrent Layer

Multiple deterrent mechanisms activated
upon bird detection

Sound, light, visual decoys, and flight
maneuvers

The system integrates these layers to create a robust solution that automatically detects and deters birds while maintaining flight stability in challenging environmental conditions.

Operational Flow



Bird Detection: AI-Powered Visual Recognition

1

Camera Input

Continuous video capture at 10-15 fps
640×480 resolution optimal for processing speed
Weatherproof housing with IP65 rating

2

YOLOv4-tiny Processing

Lightweight neural network (23 MB model size)
Runs efficiently on Raspberry Pi 4
Real-time object detection and classification
Trained specifically for bird species recognition

3

Threat Assessment

Bird proximity and approach vector analysis
Confidence threshold of ≥ 0.6 to trigger response
Tracking of bird movement across multiple frames
Classification of bird type to customize deterrent response

The YOLOv4-tiny model provides an optimal balance between detection accuracy and processing speed, allowing the system to detect birds at sufficient distance for effective deterrent activation. The model can be fine-tuned with local bird species data to improve recognition accuracy.

Vision-Based Detection

Our detection system utilizes a forward-facing camera coupled with an onboard AI model to identify approaching birds in real-time. This enables proactive deterrence before birds come within striking distance of the drone.

Hardware Components

- Raspberry Pi 4B (4GB/8GB) as main inference board
- USB Coral TPU or Intel NCS2 as edge accelerator
- CSI or USB camera module with weatherproof housing
- Total weight: approximately 170g

Software Implementation

- Raspberry Pi OS 64-bit (Lite version)
- OpenCV + Python 3.9 for image processing
- TensorFlow Lite or PyTorch Lite for model inference
- Pre-trained YOLOv4-tiny weights (custom trained on bird dataset)

Image Acquisition

Camera module captures video frames at 10-15 fps, maintaining balance between detection quality and processing requirements.

Preprocessing

Frames are resized to 416×416 pixels and normalized for optimal inference performance on the neural network.

Inference

YOLOv4-tiny processes each frame to detect and classify birds with bounding boxes and confidence scores.

Threat Analysis

System analyzes bounding box position, size changes across frames, and approach trajectory to determine threat level.

Response Trigger

When threat criteria are met, system activates appropriate deterrence mechanisms based on bird proximity and behavior.

YOLOv4-tiny Detection Logic

We selected YOLOv4-tiny for its optimal balance between speed and accuracy, making it ideal for the resource-constrained environment of a drone. The model has been proven effective for bird detection in similar applications.

Detection Workflow

1. Camera continuously captures frames at 640×480 resolution (10-15 fps)
2. Each frame is preprocessed and fed into the YOLOv4-tiny model
3. Model splits image into grid cells and makes predictions for each cell
4. If a "bird" is detected with confidence ≥ 0.6 , system flags it
5. Bounding box tracking determines if bird is approaching the drone
6. When bird is detected within critical proximity (approximately 30m), deterrence subsystem is triggered

 YOLOv4-tiny is approximately 2x faster than YOLOv3-tiny with slightly better accuracy, allowing real-time detection even with the limited computational resources available on the drone.

Deterrent Methods: Multi-Modal Approach

Acoustic Deterrents

- 20-40W Class-D audio amplifier with 100+ dB speaker
- Predator calls (hawk/eagle screeches)
- Distress calls specific to local birds
- Randomized sounds to prevent habituation

Predator Decoy

- Lightweight 3D-printed hawk/falcon head (~100g)
- Front-facing mount near camera
- Reflective eyes to enhance deterrent effect
- Exploits birds' innate fear of predators

Visual Strobe Deterrents

- High-intensity LED array with capacitor-driven circuit
- 10-20 Hz flash rate to disorient approaching birds
- Alternating colors (white, blue, red) for maximum effect
- Weatherproof epoxy-sealed dome housing

Evasive Flight Maneuvers

- Sudden vertical climb (confuses attacking raptors)
- Rapid zig-zag patterns
- 90° yaw-spin movements
- Triggered when birds approach at high speed

Multiple deterrent methods are employed simultaneously to create an overwhelming multi-sensory response that exploits birds' natural fear responses while preventing habituation to any single deterrent.

Control Flow: Prioritized Command Chain



Bird Detection

Camera feed processed by YOLOv4-tiny
Bird identified with confidence score ≥ 0.6
Proximity and approach vector calculated



Threat Assessment

Bird size and species classification
Distance and approach speed evaluation
Collision risk calculation



Deterrent Activation

Multi-modal deterrents triggered
Sound, light, and visual deterrents activated
Evasive maneuver command sent to flight controller



Response Monitoring

Continued tracking of bird position
Deterrent effectiveness evaluation
Escalation if initial response insufficient

Autonomous Decision Hierarchy:

1. Bird cleared → resume mission
2. Low battery → Return to Home (RTH)
3. Persistent bird threat → manual override via LoRa
4. GPS failure → hover using optical/IMU fusion

The system employs a structured decision tree to ensure appropriate responses to various scenarios, with autonomous operations as primary and human override as backup for complex situations.

Autonomous & Control Architecture

Dual-Layer Control System

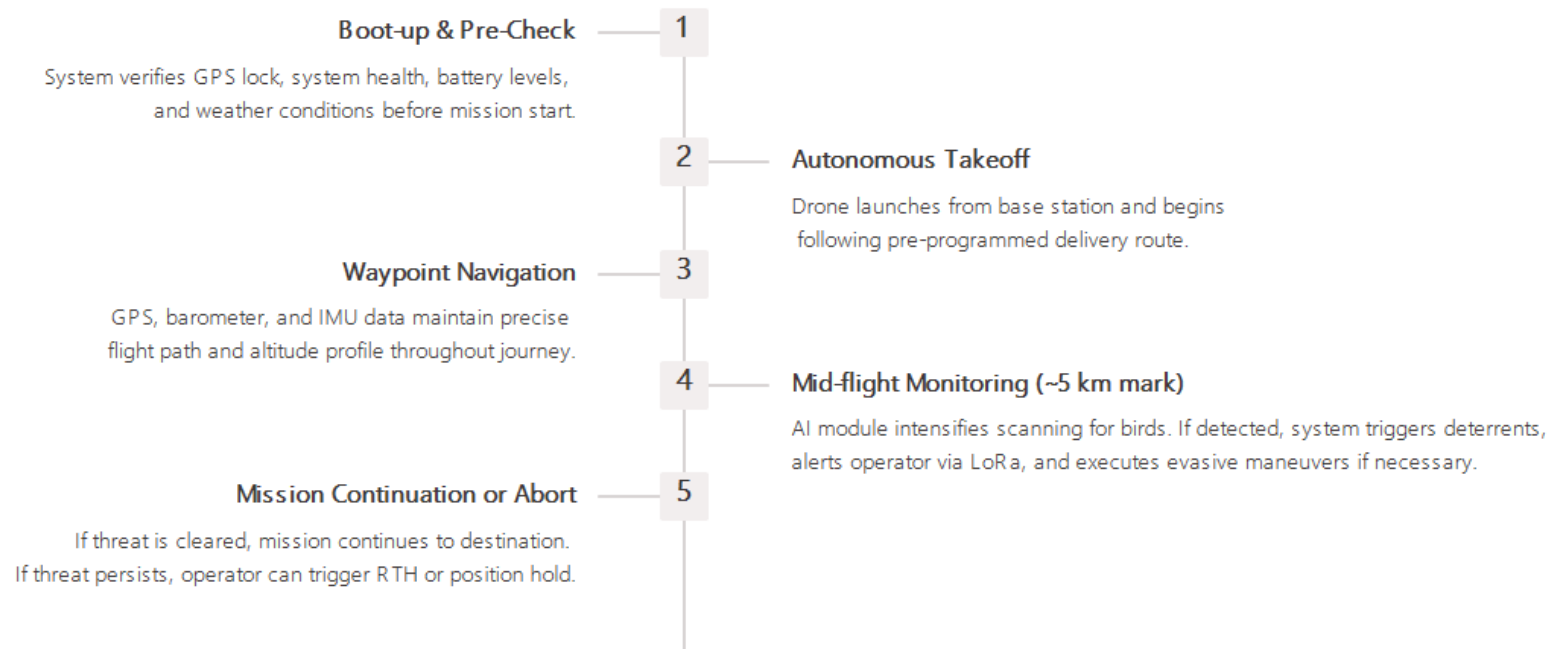
Layer 1: Autonomous Operation (Primary)

- Flight Controller (Pixhawk/ArduPilot) manages motors, ESCs, and executes flight path
- Companion Computer (Raspberry Pi) runs bird detection AI and deterrent logic
- Sensors including IMU, GPS, barometer, and optional LiDAR/stereo camera
- Geo-fence, low battery RTH, loss of GPS hover/land, and attitude safety failsafes

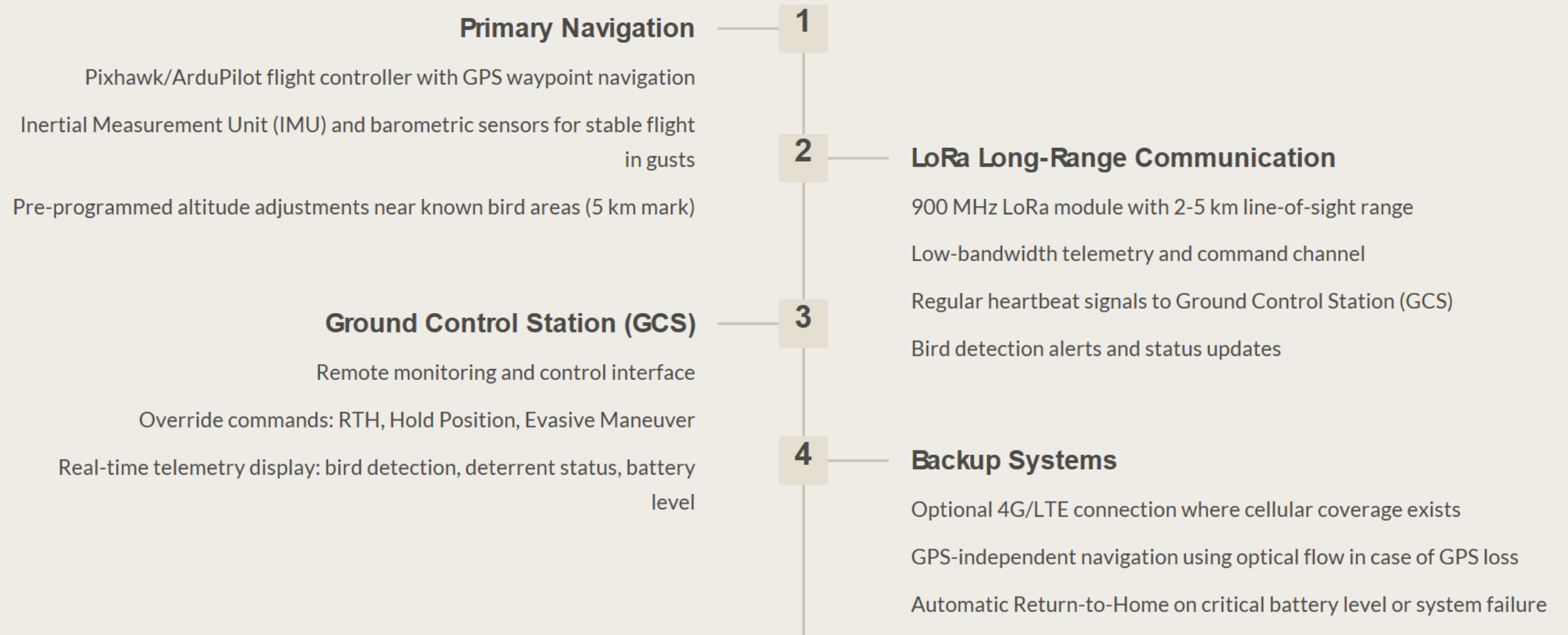
Layer 2: Remote Assist (Secondary/Failsafe)

- LoRa (900 MHz) Module for long-range command/telemetry (2-5 km LOS)
- Optional LTE/4G Dongle as secondary link where mobile signal exists
- Ground Control Station (GCS) for receiving alerts and issuing manual overrides
- Ability to command RTH, position hold, or evasive maneuvers remotely

Autonomous Flight Workflow



Communication & Failsafe Systems



Multiple communication channels and failsafe mechanisms ensure reliable operation even in challenging conditions or when primary systems are compromised.

Weatherproofing Strategy

Our comprehensive weatherproofing approach protects all critical components while maintaining system functionality:

Component	Protection Strategy
Electronics	IP65 housing + silica gel + conformal coating
Motors	Brushless IP54 or higher rated
Speaker	Horn-type with protective mesh + silicon seal
Strobe LED	Waterproof epoxy-sealed dome enclosure
Predator Decoy	Waterproof plastic with UV-resistant paint
Camera	IP67 rated housing with clear lens cover
Connections	Waterproof connectors with silicon gaskets

Signal Reliability

To maintain communication in challenging terrain:

- Redundant communication systems (LoRa + optional 4G/LTE)
- Enhanced GPS module with backup inertial navigation
- Algorithms to handle temporary signal loss without mission failure

Weight Budget Breakdown

Total Weight: 1.0 kg (Under 1.5 kg limit)



Bird Detection Subsystem: 170g

- Raspberry Pi 4 Model B: 46g
- Coral USB Accelerator/Intel NCS2: 20g
- Pi Camera v2 (8MP) or Arducam IMX519: 10g
- MicroSD Card + Case + Heatsinks: 10g
- 3D-printed camera mount (PETG): 15g
- Wiring, GPIO headers: 5g
- Thermal padding and miscellaneous: 64g



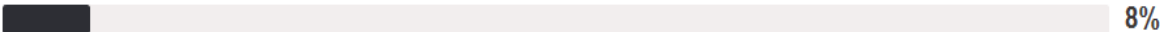
Communication Failsafe Subsystem: 110g

- LoRa SX1278 module + antenna: 35g
- Interface PCB (UART/I2C mux): 15g
- External GPS module (redundancy): 30g
- Antenna mounts + shielded casing: 30g



Deterrence Mechanism Subsystem: 640g

- Mini speaker module (DFPlayer + 3W speaker): 50g
- Strobe LEDs + driver circuit: 30g
- Foldable plastic eagle/kite decoy with servo: 80g
- Servo motor (SG90 or MG90S): 20g
- Custom 3D-printed mounts and arms: 60g
- Control PCB/relay board: 60g
- Cabling (JST/dupont wires): 20g
- Dedicated LiPo battery (2S 1000mAh): 150g
- Weatherproof casings and shielding: 50g
- Additional components and fasteners: 120g

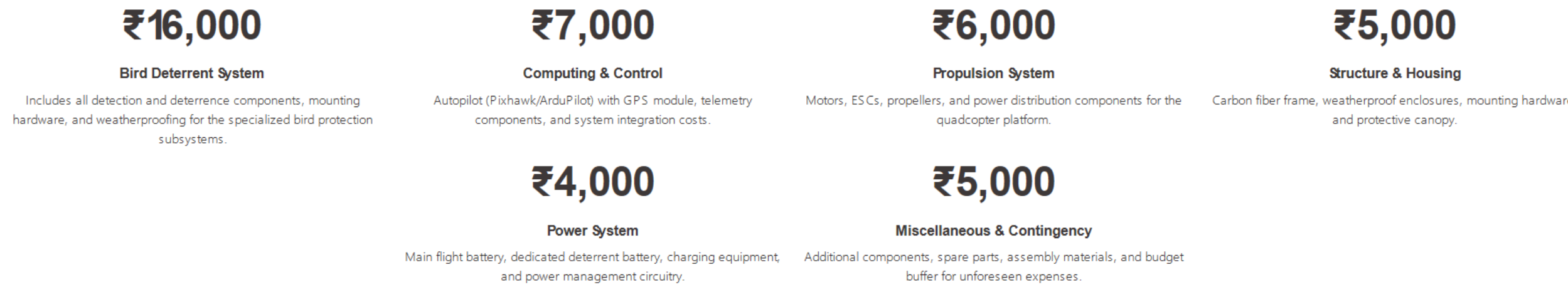


Miscellaneous: 80g

- Mounting hardware (bolts, vibration dampeners): 30g
- Design buffer for redundancy: 50g

Cost Budget Breakdown

Total Cost: ₹43,000 - ₹47,000 (Under ₹50,000 limit)



Detailed Bird Deterrent System Cost Breakdown

Component	Quantity	Cost (₹)
Camera module (1080p CMOS)	1	1,000
Raspberry Pi 4 (with heatsink)	1	4,000
Speaker + amplifier system	1	3,000
Strobe LED assembly	1	2,000
Predator decoy (plastic)	1	1,000
LoRa radio + antenna	1	2,000
Dedicated power system	1	2,000
Miscellaneous (wiring, mounts, housing)	Various	1,000
Total		16,000

The remaining budget (approximately Rs 27,000 – 31,000) is allocated to the base drone platform components, including flight controller, motors, frame, batteries, and weatherproofing. This allocation ensures a high-quality platform capable of reliable medicine delivery while maintaining sufficient budget for the specialized bird deterrent systems.

With a total estimated cost between ₹43,000 and ₹47,000, the system remains comfortably under the ₹50,000 budget limit, allowing for potential minor cost overruns or the addition of enhanced features if budget permits.

Conclusion & Benefits

Key Advantages

- Modular design allows for easy maintenance and component replacement
- Multi-modal approach prevents bird habituation to any single deterrent
- Lightweight system (1 kg) with 33% weight buffer for future enhancements
- Weatherproof design suitable for challenging hilly environments
- Autonomous operation with human override capability for complex scenarios

Real-World Applications

- Medical supply delivery in remote hilly regions
- Expandable to other drone delivery applications in areas with bird hazards
- Adaptable to different environments by modifying deterrent strategies
- System architecture can be scaled for larger or smaller drones

✔ **Mission Success:** The designed system fulfills all requirements while providing reliable protection against bird attacks. The multi-layered approach ensures redundancy and adaptability to various scenarios, allowing the drone to complete its critical medical delivery mission in challenging environments.

Our bird-deterrent system combines cutting-edge AI detection with proven deterrent methods in a lightweight, cost-effective package that ensures safe and reliable medicine delivery in hilly regions despite avian threats.